

Path 1: Pre Security > Module 4: Computer Fundamentals > Room 1: Inside a Computer System

<https://tryhackme.com/room/insideacomputer>

Inside a Computer System:

This room covers the basic components of a computer system.

>> Task 1: Introduction

Welcome to the first room of the Computer Fundamentals module! Before we can talk about security, we need to first understand what we are securing. Consider the following analogy:

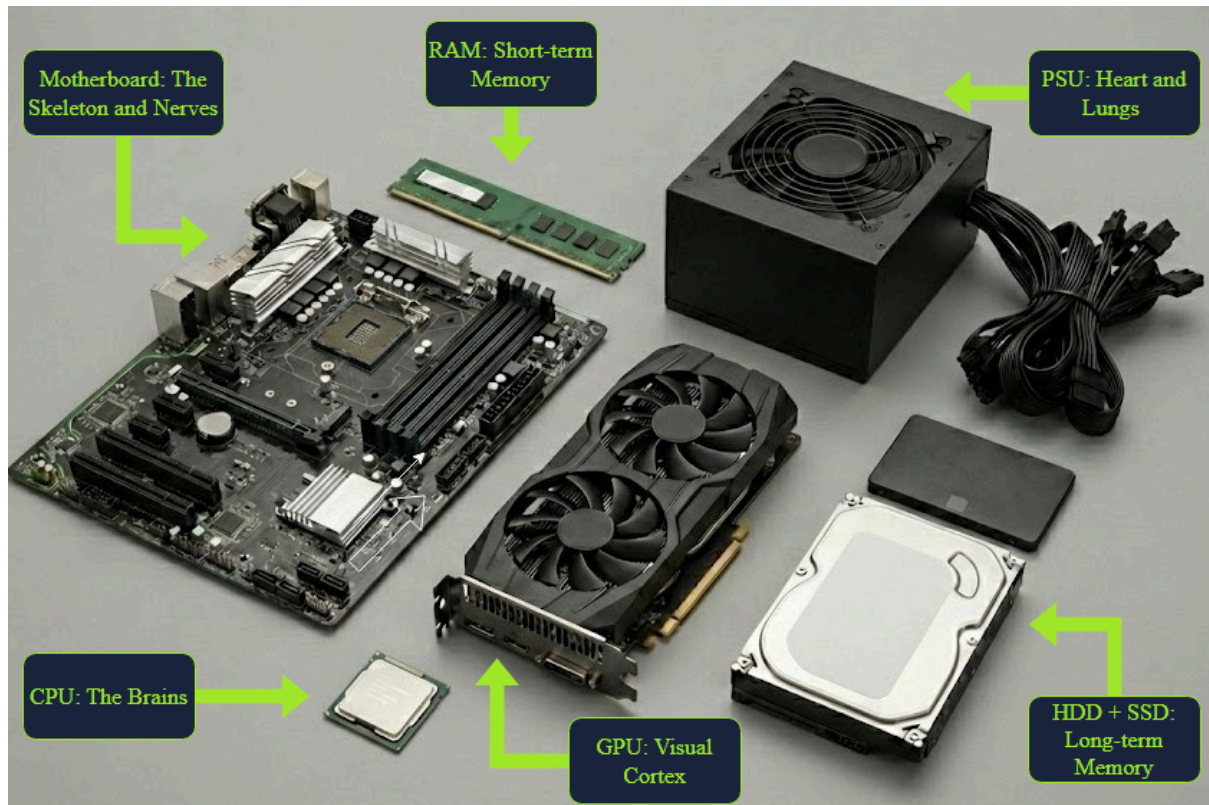
Before protecting a castle, we need to know the layout of the castle: where the treasure room, the food storage, and the commander's quarters are. We need to know who enters and how they enter the castle. Also important is who can enter these rooms and take or put treasure or food, for example.

The bottom line is: Trying to defend what you don't understand is like defending a castle you have never seen.

In this room, we will explore our "castle". We will cover what a computer is, its building blocks, and how they are connected. After completing this room, you will have a general idea of how the components of a computer system interact with each other to provide services to its users.

>> Task 2: Inside a Computer System

Nearly every computer system that you can think of includes, in one way or another, the same building blocks. Each part has its own job, and together they make the computer work. Let's have a look at each of these building blocks.



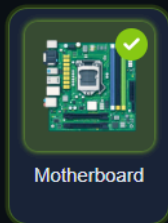
Throughout this task, we will use a very relatable analogy: the human body. The image above shows the different PC components and its human counterpart. To learn more about each component, open the "Static Site" by clicking on the green "View Site" button below. The static site should open in split-screen view.



Click a component to discover what it does. Complete all 8 to unlock the next section.

Progress

8 / 8



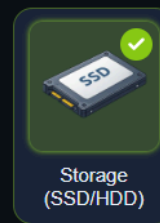
Motherboard



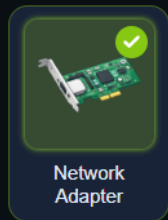
CPU



RAM



Storage
(SSD/HDD)



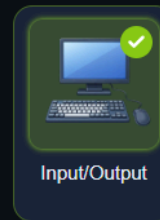
Network
Adapter



Power Supply
(PSU)



Graphics Card

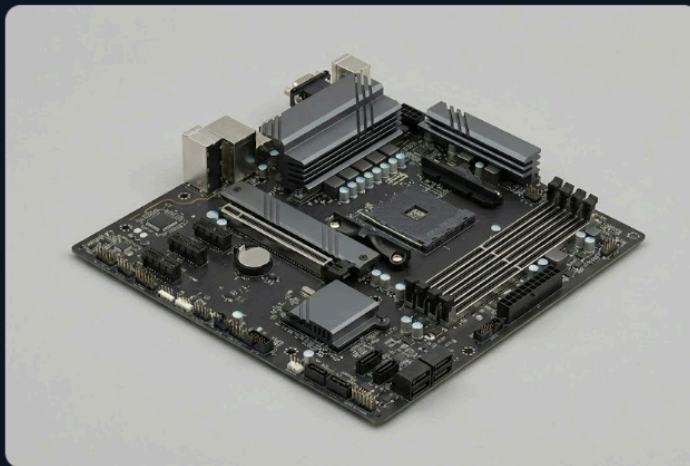


Input/Output

Motherboard



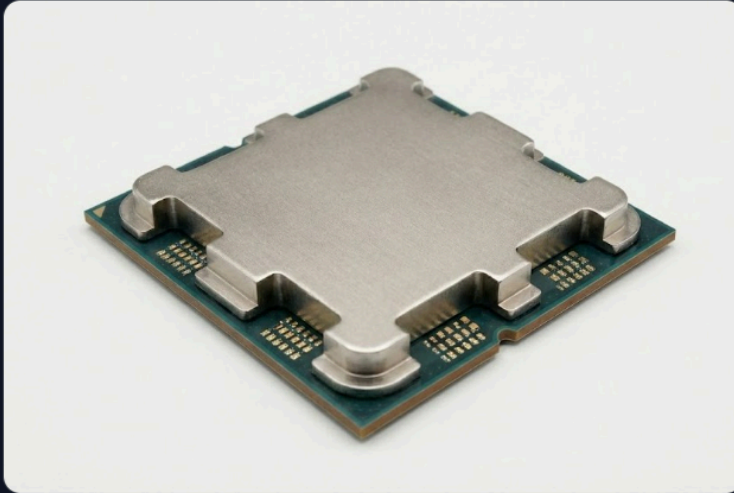
The motherboard is like our body's skeleton and nervous system. It holds all the different components in place and connects them. On a typical desktop motherboard, you'll see different connectors that house all your components - CPU socket, RAM slots, expansion slots, and various ports. Every other component plugs into or connects through the motherboard. The image below shows a typical desktop motherboard.



CPU



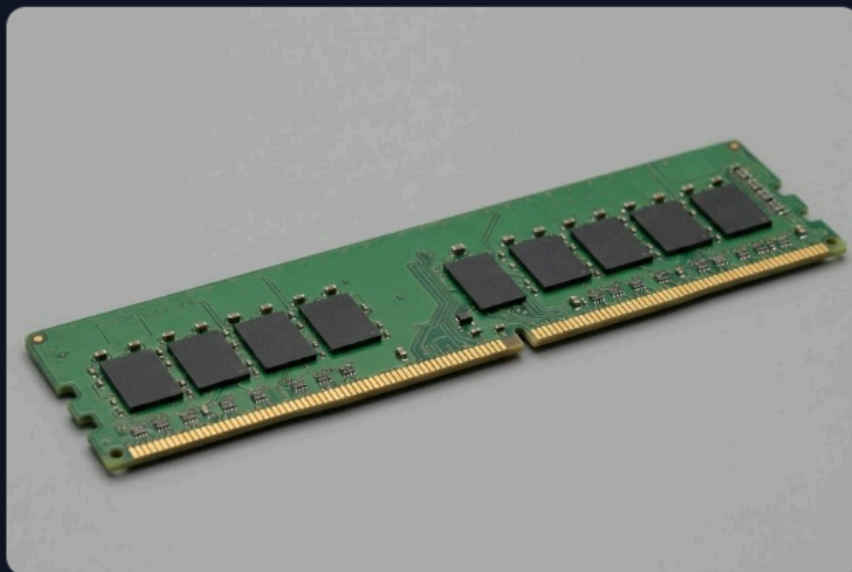
The CPU (Central Processing Unit), often called the processor, is comparable to a part of our brain. Just like our brain continuously executes instructions (add numbers, pour milk in a bowl, and so on), a CPU does the same for a computer. Modern CPUs have multiple cores that handle instructions in parallel. The CPU connects to the motherboard via the CPU socket. The image below shows a typical CPU for a desktop.



RAM



RAM (Random Access Memory) is comparable to our brain's short-term or working memory. When working on a task, we keep relevant information in mind temporarily. RAM does the same - it holds data that the CPU needs quick access to. RAM is volatile: when power is lost, all contents are gone. Modern RAM modules use technologies like DDR5 or DDR6 for increased speed and performance. The image below shows a RAM module for a desktop.



Storage (SSD/HDD)



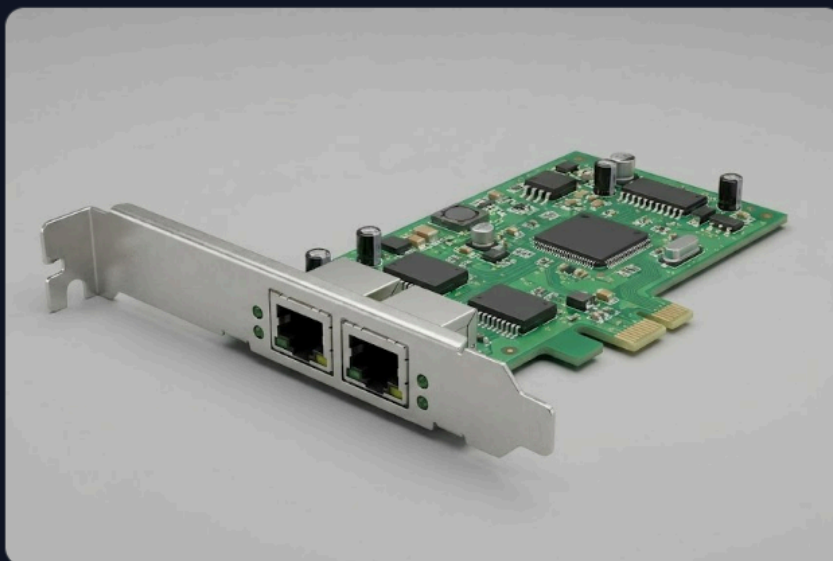
SSDs and HDDs are storage devices, comparable to our long-term memory. Just like fond memories are remembered permanently, data is saved long-term on storage devices. HDDs use older technology with moving parts, limiting performance. SSDs have no moving parts and use memory chips, allowing much faster speeds. HDDs remain popular for their large capacity at low cost. Storage connects via SATA cables or PCI Express slots. The image below shows an HDD on the left-hand side and an SSD on the right-hand side.



Network Adapter



Just like we use our vocal cords to communicate with our environment, a network adapter lets computers communicate with other systems. Network adapters come in wireless and wired variants. Often they're embedded in the motherboard, but they can also be added as expansion cards. Network cards typically connect via PCI Express ports. The image below shows a plugin network card, typically used in desktops.



Power Supply (PSU)



Every system needs power. Just as our heart pumps blood to our organs, a PSU supplies energy to all system components. The PSU is essential and requires careful consideration - if components need more power than the PSU can provide, the system will fail. The PSU takes power from an outlet and distributes it via various connectors like the main motherboard connector and Molex connectors. The image below shows a PSU used in desktops.



Graphics Card



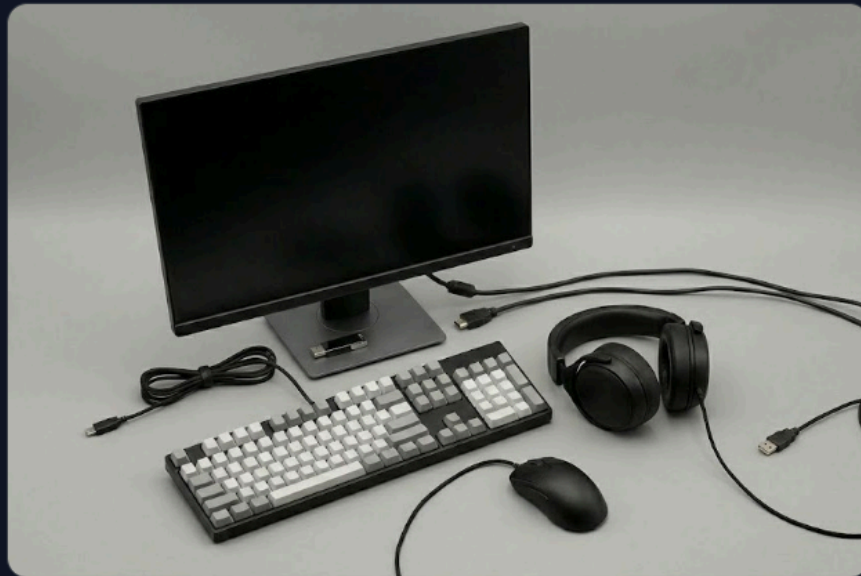
The graphics card is comparable to the visual cortex of our brain. Our eyes pick up information and the visual cortex processes it into images. Similarly, the graphics card receives information from the operating system and programs, then outputs processed visual data to a monitor. Graphics cards connect to PCI Express slots on the motherboard. The image below shows a modern day graphics card used in desktops.



Input/Output



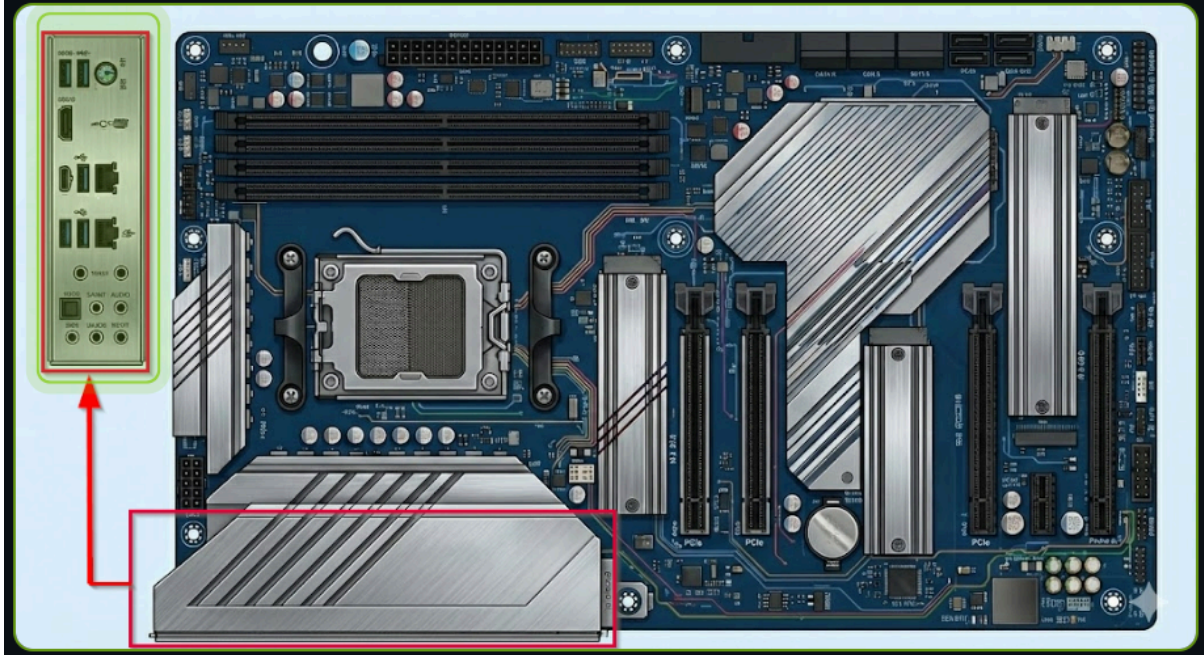
Just like we have senses to obtain information for our brains to process and then act on, computers have input and output devices. Input devices include keyboards, microphones, mice, and scanners. Output devices include monitors, printers, and speakers. Common connectors for these peripherals include USB, HDMI, and DisplayPort. The image below shows a few of the basic I/O devices



MotherBoard Inspection:

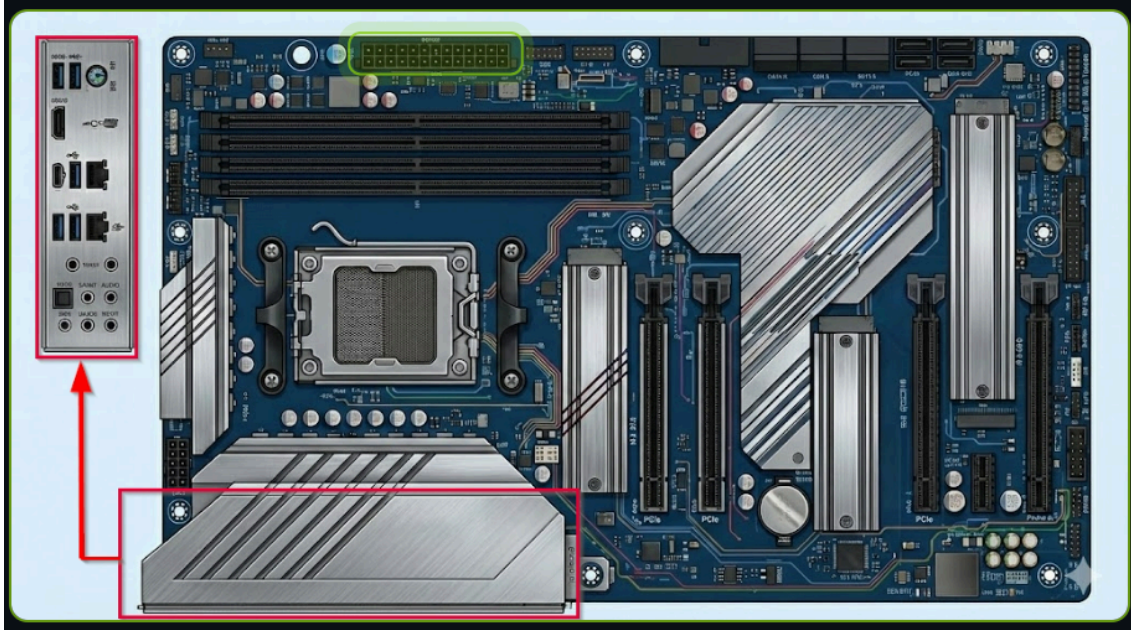
Rear I/O ports — connects the **Keyboards, mice, monitors and USB devices**.

Contains USB, network, and video ports used to connect external input and output devices.



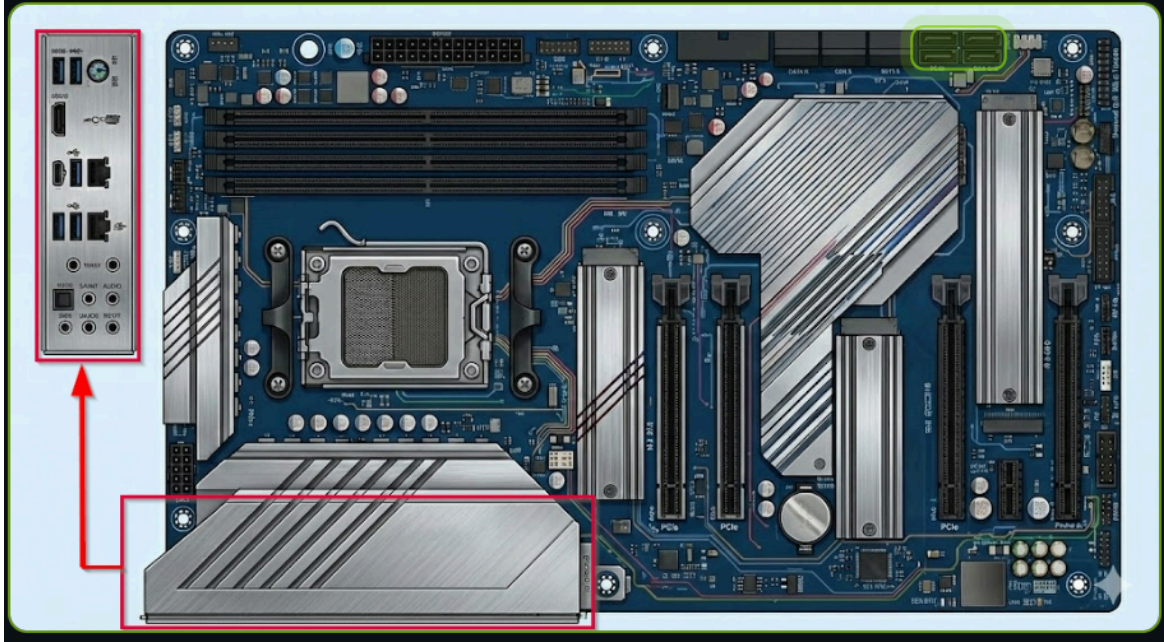
24-pin ATX power connector — connects the **Power supply unit (PSU)**.

Main power connector from the PSU that provides power to the motherboard and many components.



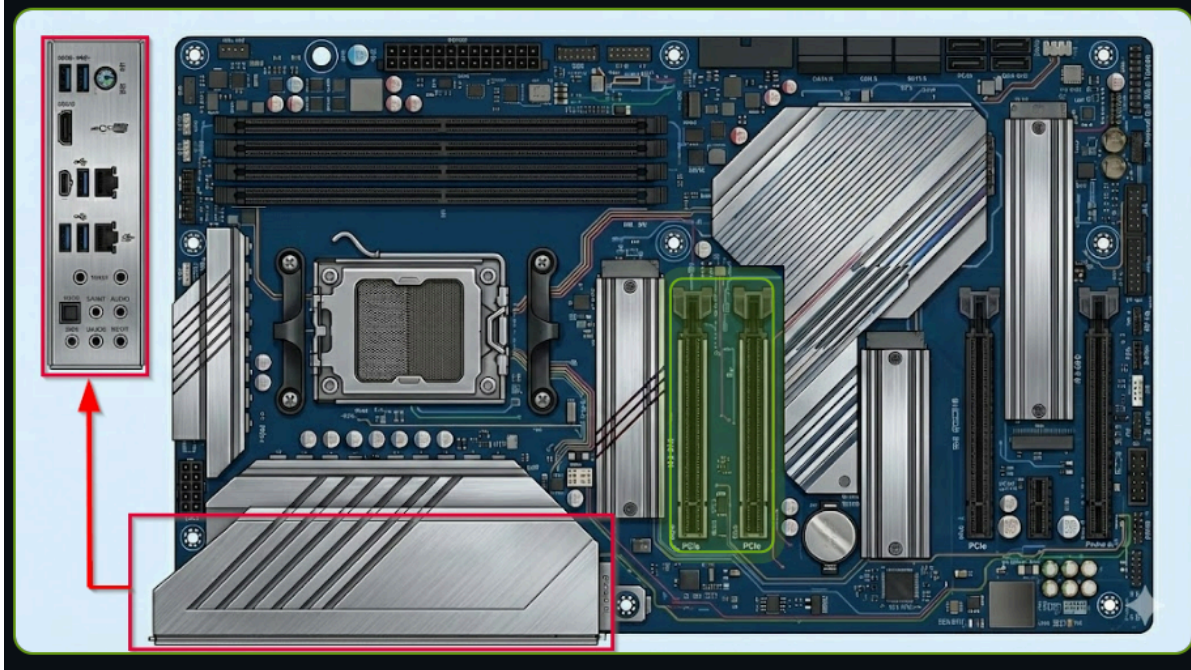
SATA ports — connects the 2.5" / 3.5" SSDs and HDDs.

Thin L-shaped data connectors used to attach SSDs and hard drives via SATA cables.



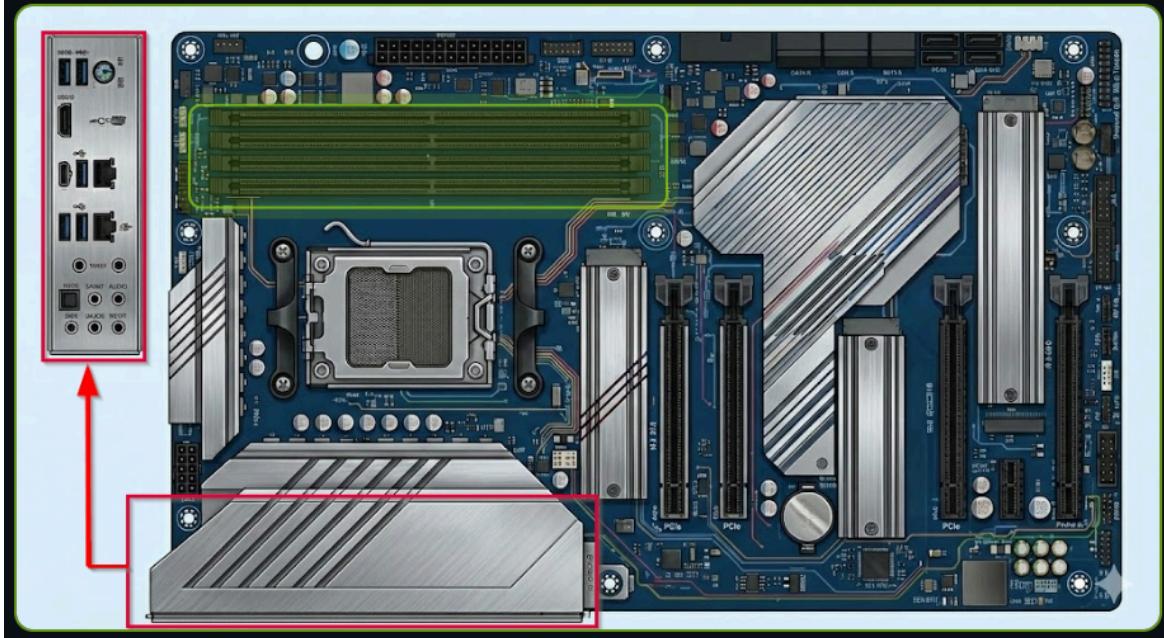
PCI Express x16 slot — connects the Graphics card (GPU) / Network Card.

The long reinforced slot is typically used for the graphics card, but can also host other high-bandwidth cards.



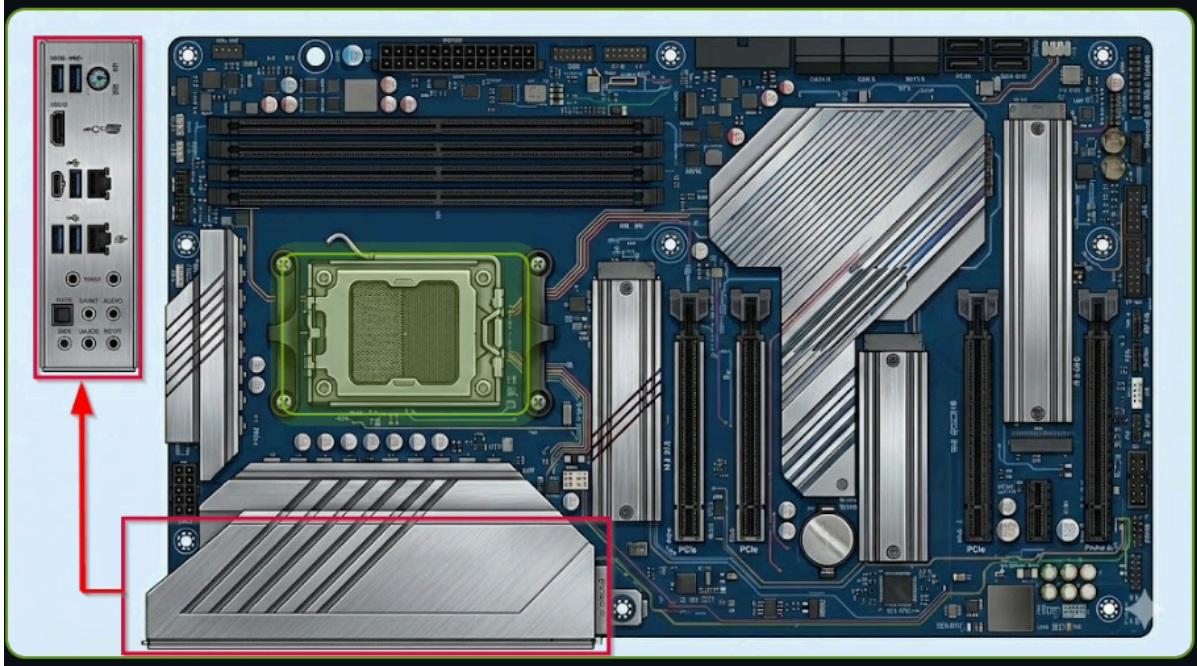
RAM slots (DIMM) — connects the **RAM modules**.

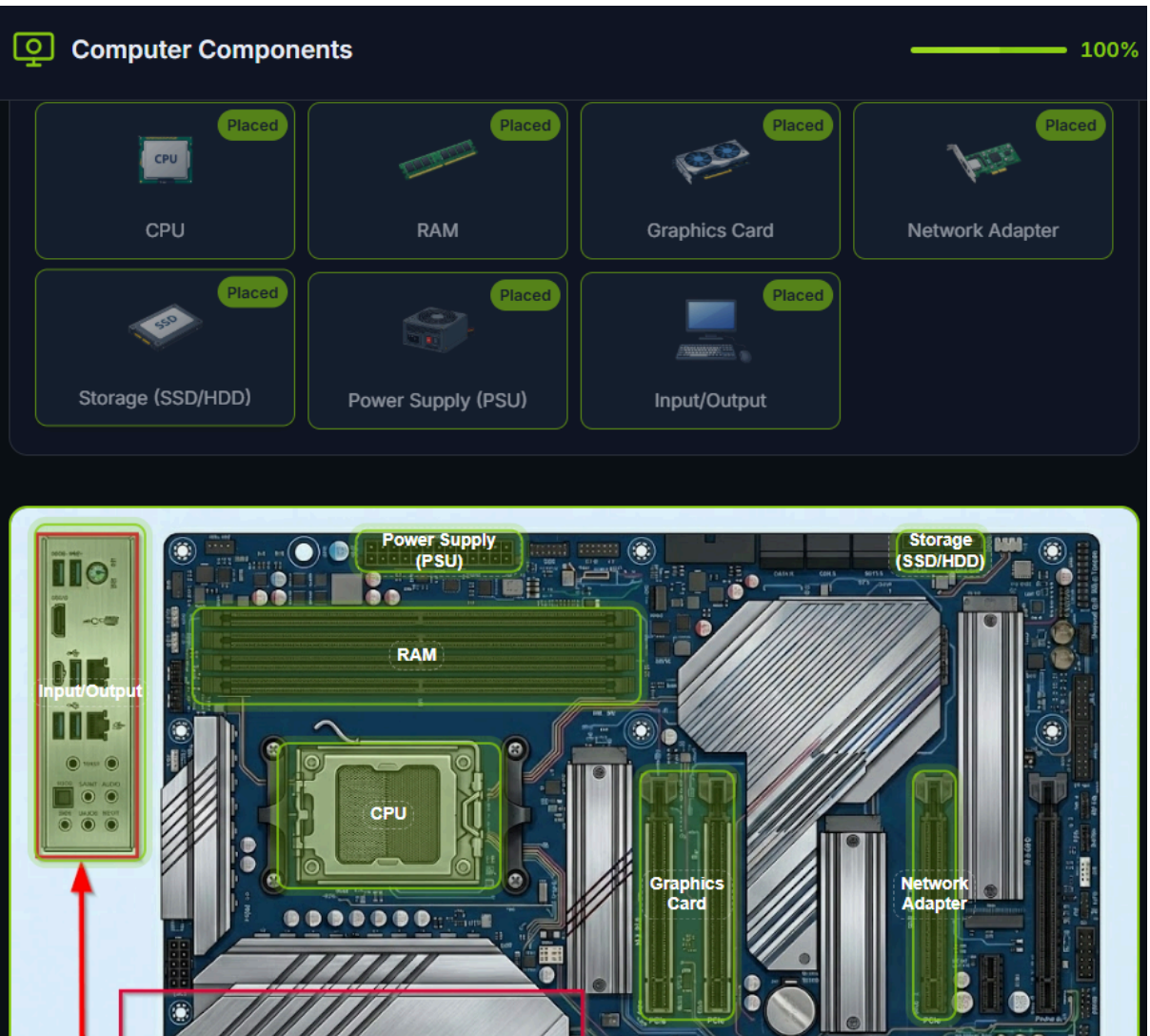
Long vertical slots where memory modules are installed. Most systems require matching pairs.



CPU socket — connects the **CPU (processor)**.

This large square socket holds the CPU. A small lever locks the chip firmly in place.





Q1) Give in the flag you received after completing the exercise on the static site.
Answer: THM{4llpccomp0n3nts1d3nt1f13d}

>> Task 3: What Happens When You Press the Start Button?

Now that the core components are installed in the computer system, it is time to boot up the system. We can compare this to how we wake up in the morning and do a quick check to see if everything is working. Only when everything is OK, do we get up and start our day. The image below shows the steps a computer system goes through before it shows you a working interface (in the form of an Operating System).



Step 1: Press the Power Button

When we press the power button on our computer system, a signal is sent to the PSU to allow power to flow. Imagine our body being powered off when we sleep. Once we wake up and receive oxygen, our body starts pumping blood and boots up.

Step 2: Firmware starts

Continuing our analogy from step 1, once the body has started up, our core components are up and running, but our brain is not yet conscious. Like our bodies, a computer system contains firmware that allows all its components to start up. The central system that manages this is called the Unified Extensible Firmware Interface (UEFI). Note: We will often see the term BIOS mentioned instead of UEFI. BIOS does the same as UEFI, but has mainly been replaced by UEFI.

Step 3: Power-On Self Test

Now that our body is up and running, it is time to test if everything is functioning as it should. If something isn't, there will be some alarm signals. One of the routines that the UEFI loads is the Power-On Self Test, which tests if every required component is present, configured correctly, and functioning.

Step 4: Select Boot Device

Once our body is up and running, configured correctly and fully functional, our system searches for the location of our bootup routine to start our consciousness. In our computer system the UEFI holds an ordered list which prioritizes on which device to look first for the boot up routine for the Operating System.

Step 5: Initiate Bootloader

Now that our system knows the part of our brain where our consciousness is located, it initiates the "load routine" to start it. Our computer systems follow a similar process: On the selected boot device, the bootloader is initiated. This bootloader transfers the Operating System from the selected boot device to the Random Access

Memory. Once the OS is transferred, the UEFI gives control over the different components to the OS.

 **Boot Sequence** 80%

Drag & Drop Challenge

Arrange the boot sequence steps in the correct order. Drag items to reorder them.

1

Press Power Button
You press the power button, sending a signal to the PSU to let power through. The system begins powering up.

2

Firmware Starts (UEFI/BIOS)
UEFI (or older BIOS) firmware initializes and coordinates components. It's the first software that runs.

3

Power-On Self Test (POST)
UEFI runs POST to check that required hardware is present, configured, and functioning. Errors trigger beeps or alerts.

4

Select Boot Device
UEFI follows a priority list to determine which device to boot from - typically an SSD or HDD with the OS installed.

5

Initiate Bootloader
The bootloader loads the operating system into RAM. Then UEFI hands control to the OS, completing the boot sequence.

Q1) What is the flag that you received after completing the exercise?

Answer: THM{pc5ucce55fully5t4rt3d}